A Gymnasium Substrate and World-Model Baseline for ARC-AGI-3:
DreamerV3 Fits the World but Cannot Yet Act in It

Hasaan Ahmad | Supervisor: Soumya Banerjee (University of Cambridge)

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1 Motivation and gap

ARC-AGI-3 is a benchmark of 25 public interactive abstract-reasoning games in which the rules and goals are *not* given and must be inferred through active play. It is scored by RHAE (Relative Human Action Efficiency): per level, $s_i = \min\big((\text{human\_actions}/\text{ai\_actions})^2,\ 1.15\big)$; a game score is the level-index-weighted mean $\sum_i s_i\, i / \sum_i i$ over all levels (including uncompleted ones), and the total score averages games equally. The ARC-AGI living survey [2] (∼80 papers across ARC-AGI-1/2/3) documents a consistent ∼3× performance cliff into ARC-AGI-3, finds only 3 of 80 papers report any ARC-AGI-3 result — *none* world-model based — and names world-model induction as the required next step. Ying et al. [3] argue the same from first principles: current evaluations test static pre-trained representations, whereas the capability of interest is using an abstract world model as a prior to build task-specific ones on the fly, for which ARC-AGI-3-style games are the proposed testbed. Lee et al. [1] give the one direct precedent — DreamerV3 beating PPO on multi-step ARC-AGI-1 — and call for scaling to harder interactive tasks. This work is the first model-based RL (MBRL) entry on ARC-AGI-3 and supplies the missing standard RL entry point.

2 Contributions

(1) Substrate (primary artifact). The first Gymnasium- and DreamerV3-`embodied`-compatible wrapper for ARC-AGI-3: a 4102-way flat discrete action space (parameter-less `ACTION1--5`; `ACTION6` over the 64×64 click grid; `ACTION7`/undo) with per-game boolean masking on the actor logits; a $(64, 64, 3)$ `uint8` image observation in DreamerV3's native modality; an offline loader for the 340-replay human corpus; and a post-hoc RHAE harness. Action coding, loader, and RHAE are property-/regression-tested (318-test suite green). DreamerV3 is used *stock and unforked* (`size12m`, the ARC-AGI-1-precedented choice [1]), registered as an `embodied` environment with no upstream modification.

(2) Controlled negative result. A paired baseline of stock DreamerV3 on 6 public games × 2 seeds, from-scratch vs. warm-started from a cross-game world model pretrained on all 340 replays (Regime B: shared WM, per-game actor). Cross-game pretraining yields no measurable benefit within seed variance.

(3) Mechanistic diagnosis. World-model reconstruction and dynamics losses collapse to convergence while RHAE stays ≈0: the model learns to predict the environment, but the controller cannot exploit it in imagination within the budget.

3 Method

The world model (encoder + RSSM + decoders + reward/continue heads) was pretrained self-supervised on the full mixed 340-replay buffer (1×A100; image-reconstruction loss 53.66 → 0.14). Per game, the pretrained WM initialises a run with a fresh actor and critic, the game's human replays pre-populate the buffer, and online interaction proceeds for 500k env-steps at default DreamerV3 hyperparameters; the cold arm is identical but for removing the warm-start, making it the *only* difference between arms — a clean ablation of the Regime-B premise. RHAE human baselines use the upper-median of first-time-player action counts, dropping levels with < 2 completers from both numerator and denominator (coverage 70.5%, 129/183 levels).

4 Results

Only vc33 is ever non-zero, and there the per-seed sign of Δ disagrees (s0 favours warm by 0.014, s1 favours cold by 0.002): at $n = 2$ the warm−cold delta lies within seed variance, so the Regime-B premise that cross-game pretraining aids downstream control is *not supported*. The pre-registered gate (RHAE > 0 on $\geq 2/3$ of $\{\text{vc33}, \text{sb26}, \text{cd82}\}$) failed 1/3 — a clean, paired, 6-game negative result, not one unlucky run.

Table 2 is the central finding in one view. cd82's world model fits *tighter* than vc33's (image loss 0.06 vs. 0.16) — yet vc33 is the only game the controller ever solves, while cd82's policy never leaves a 1.000 random-action fraction across the entire 500k-step run. The model learns the environment; the controller cannot exploit it. This holds across all six pilot warm cells ({vc33, sb26, cd82}×2 seeds): image loss collapses >95% everywhere, while both gate-failure games clear 0/24 eval episodes in every cell. The bottleneck is the controller, not the world model — not a single lucky or unlucky run.

Table 1: Paired RHAE @ 500,000 env-steps. Stock DreamerV3 `size12m`, 6 public games $\times$ 2 seeds, cold (from-scratch) vs. warm (cross-game WM-pretrained). Δ = warm−cold. Higher is better; human-parity $\approx$ the per-level 1.15 cap.

	Cold		Warm			
Game	s0	s1	s0	s1	Cold $\bar{x}$	Warm $\bar{x}$
vc33	0.0411	0.0182	0.0548	0.0166	0.0297	0.0357
sb26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
cd82	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
tn36	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ls20	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
lf52	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Combined (mean of 6)					**0.0049**	**0.0059**

Table 2: The dissociation: the world model fits *better* on a game the controller never solves (cd82) than on the only game it ever solves (vc33). Both columns are seed-0 warm runs from the same paired experiment as Table 1; the pattern holds across all 6 pilot warm cells.

Signal	vc33 (only solved)	cd82 (never solved)
WM image-recon. loss	$58.3 \rightarrow 0.16$ (−99.7%)	$31.0 \rightarrow$ **0.06** (−99.8%)
WM dynamics loss	$10.6 \rightarrow 1.24$	$11.4 \rightarrow 1.03$
Train eps. clearing $\geq$1 lvl	17/9751	0/4950
Eval eps. clearing $\geq$1 lvl	4/23	0/24
Random-action frac. @ 500k	0.750	**1.000**
Post-hoc RHAE	0.0548	0.0000

5 Discussion and positioning

This is the first MBRL data point on ARC-AGI-3 and sharpens the survey's $\sim$3$\times$ cliff [2]: a model that provably fits these environments still cannot act in them under a stock controller at a realistic budget — precisely the "static representation vs. on-the-fly task model" gap of Ying et al. [3]. It also qualifies the Lee et al. [1] precedent: DreamerV3's ARC-AGI-1 success does *not* transfer to the interactive, rule-inference regime at this scale, and cross-game pretraining does not close the gap — isolating the controller, not the world model, as the locus of difficulty. **Limitations:** a compute-bounded 500k-step budget and $N = 6 \times 2$ — a scoped claim about *stock DreamerV3 at this budget ± this pretraining*, not that MBRL cannot do ARC-AGI-3; the diagnosis carries the weight, not the headline number. **Status and planned work.** The substrate (contribution 1) is complete, tested, and reusable independent of any result; the controlled negative result and its diagnosis (2–3) are established across the full 6-game paired sweep. Three steps remain, each with a pre-specified criterion that would confirm or relocate the diagnosis. *(i) Representational probing:* linear decoders trained on the frozen RSSM latents for task-relevant variables — level identity, level-transition events, and object configuration — against a label-permuted control; above-control accuracy confirms the world model encodes the structure the controller fails to exploit, whereas a null result would relocate the bottleneck from control to representation learning. *(ii) Generative-rollout fidelity:* Fréchet Video Distance between imagined and ground-truth trajectories over the actor's planning horizon, establishing whether the model's predictions remain faithful across the rollout length on which the policy is trained. *(iii) Controller-side intervention:* a bounded ablation that raises the imagination-to-environment training ratio and steepens the entropy-annealing schedule with the world model held fixed, isolating whether the gap closes under additional policy-improvement compute alone rather than a change of model class. Steps (i)–(ii) are compute-light and operate on a frozen model; (iii) is a single bounded retraining sweep. The substrate contribution is independent of all three outcomes.

References

[1] D. Lee et al. *Reinforcement Learning and Model-Based Reasoning on the Abstraction and Reasoning Corpus.* arXiv:2408.14855, 2024.

[2] *The Abstraction and Reasoning Corpus: A Living Survey.* arXiv:2603.13372, 2026.

[3] L. Ying et al. *Instance vs. Abstract World Models.* arXiv:2507.12821, 2025.

[4] D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap. *Mastering Diverse Control Tasks through World Models.* Nature, 2025. arXiv:2301.04104.

[5] *ARC-AGI-3: Interactive Reasoning Benchmark.* arXiv:2603.24621, 2026.